

Pitch

Hook:

Good evening, my name is Sageena. My research addresses a major inefficiency in traffic engineering: traffic sensor testing is duplicated across states because results are siloed. We built an AI-powered clearinghouse that standardizes sensor test results and enables cross-state reuse and natural-language querying, reducing redundant testing and improving deployment decisions.

Specific deets:

1) Point to Motivation / Introduction

This problem matters because traffic sensors directly affect signal timing and safety, yet sensor testing remains highly fragmented. (number one issue) Despite thousands of deployed sensors and repeated evaluations across DOTs and testing facilities, no shared platform exists for standardized sensor test results—forcing agencies to re-test in isolation and driving up costs and leading to inconsistent deployment decisions.

“Beyond data silos, second key gap is the lack of a shared testing framework—similar sensors are often tested using different stages and performance measures, making results difficult to compare or reuse.”

2) Point to System Architecture / Workflow

“We addressed both gaps by designing a centralized clearinghouse and a standardized testing framework. The project defines common testing stages—such as Stage 1 Level 1, Level 2, and beyond—so facilities follow the same test plans when evaluating sensors.

So verified facilities submit results using this standardized template, which are organized and presented in a shared web interface for cross-state comparison.

3) Point to AI-Assisted Querying

To make this data usable in practice, users can query validated sensor test results in natural language. Rather than sending all data to the AI, the system first narrows the dataset using structured filters—such as sensor type, test stage, and performance metrics—and then generates concise, comparable summaries from only the relevant results.

4) Point to Results

Finally, we validated the full pipeline end-to-end using synthetic sensor test reports to demonstrate feasibility of standardized submission and AI-assisted querying.

The novelty is not only in centralizing sensor test results, but in standardizing how sensors are tested in the first place, so results from different facilities can actually be compared and reused.”

If asked::

On AI rigor

“The system does not send the full dataset to the language model. It first filters for best-matching tests using structured metadata, then generates responses from a constrained JSON subset. This preserves comparability and avoids hallucination.”

On novelty

“What’s novel is not just storage, but standardization + validation + AI-assisted retrieval in a safety-critical infrastructure context.”

On real-world deployment

“The architecture mirrors existing DOT workflows — testing facilities already generate reports; we standardize and centralize them instead of changing testing itself.”